# Fairness and Discrimination in Mechanism Design and Machine Learning

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## Introduction

### Mechanism Design

- Typically used in resource allocations settings
- Ex: "How do we allocate students to schools?"

Principal commits to allocation rule → Agents report their types and receive allocations

### Machine Learning

- Typically used in prescriptive settings
- Ex: "Is a student qualified to attend university?"

Individuals present their features → A classifier predicts their "true" label

## Fairness and Discrimination

**Fairness**

Fairness in MD typically quantified as (weighted) social welfare

**Discrimination**

**Taste-based** | **Belief-based**

Statistical Discrimination
Coordination Failure
Mis-specification

Large focus on quantitative fairness definitions

Individual fairness

Demographic parity

Equalized odds

## Lessons

Tension: fairness vs welfare

Long-term effects of fairness

Strategic agents

(Re)defining notions of fairness

Group-level diagnosis

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